TFPT — Origin Theory

The seam as a horizon, the cyclic compiler hull,
and the self-reproducing parameter-free fixed point

An [E] structural core (`v53`–`v56`) with one honestly-typed [C] interpretation

Stefan Hamann   Alessandro Rizzo

June 17, 2026 · v5.1 (rev 164)

What this document is, and is not

This note collects, in one derivation, why the two TFPT inputs leave *no free dimensionless compiler dial beyond the anchor structure and* π (the one dimensionful scale v_{geo} and the continuous transfer physics F_{transfer} remain explicitly typed, not derived), and how the entire integer skeleton, the seam constant $c_3 = \frac{1}{8\pi}$ and a built-in cyclic structure all flow from one boundary pair. **Two layers, kept strictly apart:**

- **Structural core** [E] (§1–§5): exact, machine-checked identities (`v53`–`v56`, plus `v6`/`v8`/`v3`). These stand on their own, independently of any cosmology.
- **One interpretation** [C] (§6): the *cyclic self-reproduction* reading — collapse → horizon=seam → next cycle. This is a physical hypothesis, *not* derived and *not* machine-checkable; it is offered because it gives the structural fixed point a physical selection mechanism. It is flagged [C] throughout.

Nothing in the [C] layer is sold as proven; nothing in the [E] layer depends on the [C] layer.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Contents

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the

Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

- **0.** `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
- **1.** `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
- **2.** `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
- **3.** `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
- **4.** `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
- **5.** `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$ as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading companion O (Origin Theory).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$	[E]/[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels $(U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}})$ are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\rm car} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: $v_{\rm geo}$, $G_{\rm net}$, $F_{\rm transfer}$ *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

1 The whole integer skeleton from one pair $(g_{\rm car}, N_{\rm fam}) = (5, 3)$

TFPT runs on two operational inputs: the seam constant $c_3 = \frac{1}{8\pi}$ (P1) and the carrier rank $g_{\rm car} = 5$ (P2). The boundary $\mu_4 = \{1, i, -1, -i\}$ gives the four-punctured line $\mathbb{P}^1 \setminus \mu_4$ with $H_1 = \mathbb{Z}^3$, hence the family count $N_{\rm fam} = 3$ (A_3). We show the rest of the discrete data is generated by the single pair $(g_{\rm car}, N_{\rm fam}) = (5, 3)$.

One object, two canonical invariants. The pair $(g_{\rm car}, N_{\rm fam})$ is read off the *same* four-point divisor $D = \mu_4$ (degree 4) by the two canonical functors on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg D - 1 = 3 = N_{\rm fam}$ and the function side (Riemann–Roch) $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg D + 1 = 5 = g_{\rm car}$, the latter matched by $\mathrm{rank}(D_5) = 5$ (the carrier is the $\mathfrak{so}(10)$ half-spinor). Reading the carrier as this meromorphic seam-mode space (at most simple poles on μ_4) is geometrically natural [E] in its arithmetic, though the physical identification stays [C] and does not displace the over-determined $g_{\rm car} = 5$ (Pascal/bootstrap/Lean) [`verification/v189_riemann_roch_carrier.py`].

And the function side generates D_5 itself, not just $g_{\rm car}$. The even Clifford algebra of the 5-dimensional mode space closes the chain $\mu_4 \to g_{\rm car} \to D_5$: $\Lambda^{\rm even}(C_{\rm car}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16 = 2^{g_{\rm car}-1} = \dim S^+$, which is exactly the $\mathfrak{so}(10) = D_5$ half-spinor. So the carrier is not merely “rank 5 like D_5” — it *is* the Clifford spinor of the Riemann–Roch mode space [E], with the physical identification again [C] [`verification/v197_rr_carrier_clifford_d5.py`].

The $(5, 3)$ generator and the Pythagorean mass volume [E] ([`verification/v53_compiler_core.py`])

By sum, difference and halving,

$$\mathrm{rank}\, E_8 = g_{\rm car} + N_{\rm fam} = 8, \qquad |\mathbb{Z}_2| = g_{\rm car} - N_{\rm fam} = 2, \qquad |\mu_4| = \tfrac{g_{\rm car} + N_{\rm fam}}{2} = 4,$$

and $(N_{\rm fam}, |\mu_4|, g_{\rm car}) = (3, 4, 5)$ is *the* Pythagorean triple. This is load-bearing: the grand-mass-volume exponent $\Delta_Y = g_{\rm car}^2 = 25$ ($\det M_{\rm SM} \sim (\varphi_0)^{25}$) splits as a *difference of squares*,

$$\boxed{\Delta_Y = g_{\rm car}^2 = \underbrace{N_{\rm fam}^2}_{\text{down}=9} + \underbrace{(g_{\rm car} - N_{\rm fam})(g_{\rm car} + N_{\rm fam})}_{= |\mathbb{Z}_2| \cdot \mathrm{rank}\, E_8 = \dim S^+ = 16} = 9 + 16 = 25}$$

(the K-row sums $(6, 9, 10)$: down carries $N_{\rm fam}^2$, up+lepton carry $|\mathbb{Z}_2| \cdot \mathrm{rank}\, E_8$). The *anchor* $a = (1, 1, 2)$ is the unique 3-multiset whose elementary symmetric functions are the compiler atoms $(|\mu_4|, g_{\rm car}, |\mathbb{Z}_2|) = (4, 5, 2)$; equivalently $\chi_a(t) = (t-1)^2(t-2) = t^3 - |\mu_4| t^2 + g_{\rm car} t - |\mathbb{Z}_2|$.

The two glue discriminant norms are likewise $(g_{\rm car}, N_{\rm fam})$ over the glue index: $q(D_5) = g_{\rm car}/|\mu_4| = \frac{5}{4}$, $q(A_3) = N_{\rm fam}/|\mu_4| = \frac{3}{4}$, so their sum is the E_8 root norm 2, their difference is the lepton transport value $\delta = |\mathbb{Z}_2|/|\mu_4| = \frac{1}{2}$, their product is $\dim \mathfrak{su}(4)/\dim S^+ = \frac{15}{16}$ and their ratio is $g_{\rm car}/N_{\rm fam} = \frac{5}{3}$ [`verification/v51_boundary_half_step.py`]. The single transcendental is π; both π-coefficients are 2π times a compiler integer.

The icosahedral bedrock: *why* the atoms are $\{2,3,5\}$ [E] ([`verification/v219_icosahedral_mckay.py`])

The pair $(g_{\rm car}, N_{\rm fam}) = (5,3)$ and the sheet $|\mathbb{Z}_2| = 2$ are not three free small integers: $\{2,3,5\}$ is forced as soon as one fixes E_8. By the **McKay correspondence** the finite subgroups of $SU(2)$ are classified by the affine ADE diagrams, and E_8 is the exceptional *maximum* of that spherical tower ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$). Its branch data is the *icosahedron*: the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has element orders $\{1,2,3,4,5,6,10\}$ — it literally *contains* the $2,3,5$ rotation-axis orders — and its nine irreducible representations have dimensions $\{1,2,2,3,3,4,4,5,6\}$, the Kac marks of $\hat{E}_8$, with

$$\sum_i d_i = 30 = h(E_8) = |\mathbb{Z}_2| \cdot N_{\rm fam} \cdot g_{\rm car} = 2\cdot 3\cdot 5, \qquad \sum_i d_i^2 = 120 = |R^+(E_8)|.$$

The McKay graph is *built from the group* (the natural 2-dim rep gives the affine E_8 adjacency, top eigenvalue 2, with marks = degrees), so the recurring $\{2,3,5\}$ signature has a single bedrock: E_8 *is* the icosahedron. This is a *backward certificate* of the already-closed E_8 [E]; the reading "it explains the $(2,3,5)$ choice" is [C]; the raw-seam origin (without first positing E_8) stays the [O] P2 interface. The same exceptional geometry has two metric readings via complex multiplication ([`verification/v222_cm_norm_duality.py`]): on the *square* seam ($j{=}1728$, Gaussian $\mathbb{Z}[i]$) the EM index is the Gaussian norm $N(g_{\rm car}{+}i|\mu_4|) = 5^2{+}4^2 = 41 = 10b_1$, while on the *hexagonal* partner ($j{=}0$, Eisenstein $\mathbb{Z}[\omega]$) the scalaron exponent is $N(3{+}2\omega) = 7$ — the carrier split $(3,2)$ generates $(5,6,7,13)$ by sum/product/Eisenstein/Gauss norm, the rings rigid. And the seam μ_4 clock is the order-4 character of the E_8 Coxeter cycle itself: $(\mathbb{Z}/30)^\times$ are the E_8 exponents, with the order-4 generator 7 acting as a literal μ_4 clock on the eight live phases ([`verification/v223_coxeter_totative_clock.py`]). Finally, P2 itself reads as a Riemann–Roch index: a degree-4 seam divisor on $\mathbb{P}^1$ forces $h^0{=}5{=}g_{\rm car}$ and rank $H_1{=}3{=}N_{\rm fam}$, with even Clifford closure $\Lambda^{\rm even}(\mathbb{C}^5) = 16 = \dim S^+$ ([`verification/v228_rr_index_gate.py`], [E]/[C]/[O], conditional on the seam producing the degree-4 divisor = `QGEO.SYM.01`). And the same icosahedron gives the seam a *geometric* home ([`verification/v232_e8_kleinian_seam.py`]): on the du Val side of McKay, the minimal resolution of the Kleinian singularity $\mathbb{C}^2/2I$ has an exceptional locus of *eight* rational curves glued in the E_8 pattern (their negated intersection form is the E_8 Cartan, det 1; each a (-2)-curve), and its link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ is perfect, so $H_1 = 0$, $\pi_1 = 2I$ of order $120 = |R^+(E_8)|$). The eight curves are rank $E_8 = g_{\rm car} + N_{\rm fam}$ — a *fourth* reading of the "8" as the eight seam components. So the raw seam ($\mathbb{P}^1$ with marks) is canonically one of the E_8 du Val exceptional curves: `QGEO.REALIZE.01` acquires an algebraic-geometric model [C], though it still presupposes E_8 (the bedrock stays [O]).

2 The "8" is triply forced: geometry = lattice = gravity

The integer 8 in $c_3 = \frac{1}{8\pi}$ is not a free choice. Three independent routes give it, and the seam=horizon reading requires their agreement — which holds, overdetermined. **Firewall (status of the "8").** The geometry and lattice routes are exact [E]. The gravity route is an *exact arithmetic alignment* with the Hawking/Einstein 8π normalisation, hence also [E] *as an alignment*. What is *not* [E] is the physical statement "the seam *is* a horizon": that principle stays [C] until it is derived inside the theory (the Seam–Horizon Theorem, §8). The exact alignments below are [E]; the physical identification is [C].

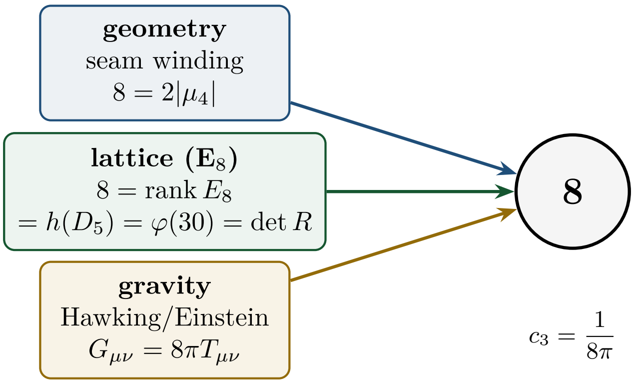

Figure 1: The seam denominator 8 is fixed three independent ways. If the seam is a horizon, the gravitational 8π (Hawking temperature $T_H = c_3/M$, Einstein $G_{\mu\nu} = 8\pi T_{\mu\nu}$) forces c_3; it must then coincide with the geometric $2|\mu_4|$ (Gauss–Bonnet seam winding) and the lattice rank E_8. All three give 8. [E] ([`verification/v54_seam_horizon_keystones.py`])

One boundary transport for flavor *and* horizon [E] ([`verification/v54_seam_horizon_keystones.py`])

The boundary transport spectrum is $\{1, (2/3)^6, (1/3)^6\}$ (cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ at the four μ_4-punctures). Its sub-leading eigenvalue $\lambda_2 = (2/3)^6$ appears in *both* sectors:

$$\text{SM flavor gap} \quad \Delta_{\text{gap}} = -\log(2/3)^6 = 6\log\tfrac{3}{2} = 2.4328$$
$$\Longleftrightarrow \qquad \text{horizon Page recovery} \quad I_n \sim \lambda_2^n = (2/3)^{6n}.$$

One boundary operator governs the Standard-Model flavor hierarchy and the horizon information recovery — exactly what "the SM is the boundary data of the seam=horizon" would require.

The maximal black hole is the anchor [E] ([`verification/v101_horizon_anchor.py`])

The seam=horizon reading now has a *classical-gravity* fingerprint with zero free parameters. Put a black hole into the de Sitter bulk (Schwarzschild–de Sitter): the maximal case (Nariai, two horizons merging) has, in units $1/\sqrt{\Lambda}$, the horizon cubic

$$\boxed{t^3 - 3t + 2 = (t-1)^2(t+2): \ \text{roots } (1, 1, -2)}$$
$$= \ \text{the } \textit{traceless projection} \text{ of the anchor } a = (1, 1, 2),$$

with coefficients $(-N_{\text{fam}}, |\mathbb{Z}_2|)$ and $M_N = 1/(N_{\text{fam}}\sqrt{\Lambda})$. Each Nariai horizon carries S_{dS}/N_{fam}; the total is $\frac{2}{3}S_{dS}$ — **the Koide branch value $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$ is the exact entropy bound of the maximal black hole**. The interpolation is $S_{\text{tot}}/S_{dS} = (x^2+1)/\Phi_3(x)$ (the N_{fam} cyclotomic), the three-root entropy total $\pi \sum r_i^2 = |\mathbb{Z}_2| S_{dS}$ is conserved for every M, the mass line is itself a split double cover (branch points $\pm 1/N_{\text{fam}}$, separation $2/3$, deck = horizon swap — the gravitational twin of the flavor sheet $\mathbb{Z}_2$), and the evaporation flow runs *away* from the anchor point toward the democratic endpoint — the same repeller/attractor orientation as the flavor relaxation (§4). Six independent landings on already-load-bearing atoms; the GR identities are textbook, the bulk reading stays [C]. Details and figures: Appendix H.

Variational upgrade ([`verification/v102_seam_orientation.py`]). The orientation is exact on both sides: the flavor relaxation is the *gradient flow* of a cubic potential whose critical points are the two branch points with stationary curvatures $\pm\Delta$ (= the transfer gap; inflection at scalaron/2; Lyapunov rate $d(-\ln\rho)/dt = \Delta$ constant), and the SdS entropy functional has the Nariai/anchor point as its *unique* stationary point with curvature $2/9 = |\mathbb{Z}_2|/N_{\text{fam}}^2$. **Both sectors flow away from an anchor-stationary configuration with grammar-constant curvatures** (Fig. 2); the "one variational principle of the seam" reading stays [C].

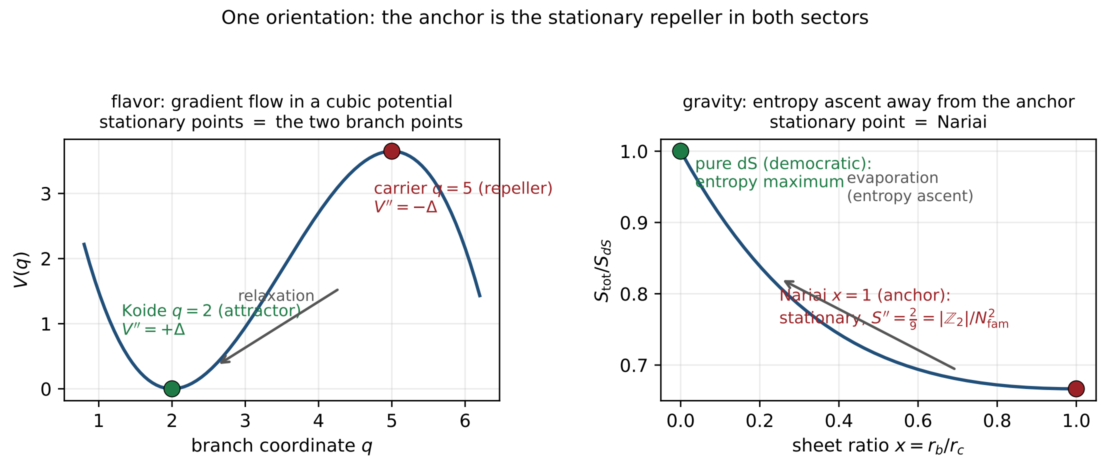

Figure 2: One orientation, two sectors: the anchor configuration is the stationary repeller of the natural functional — the cubic flavor potential (curvatures $\pm\Delta$) and the SdS entropy (curvature $2/9 = |\mathbb{Z}_2|/N_{\rm fam}^2$). **[E]** ([`verification/v102_seam_orientation.py`]); the common-principle reading stays [C].

3 A cyclic element *inside* the hull: the order-30 Coxeter rotation

The compiler hull E_8 carries a *literal* cyclic symmetry, computed from first principles.

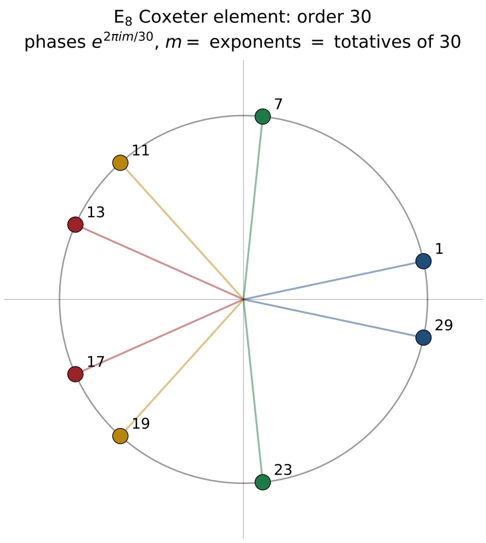

Figure 3: The E_8 Coxeter element (built from the Cartan matrix as $c = s_1 \cdots s_8$) has order exactly $h(E_8) = 30$; its eigenvalues are the primitive 30th roots $e^{2\pi i m/30}$ with m the E_8 exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$, which are precisely the $\varphi(30) = 8$ totatives of 30. The phases pair as $m + (30-m) = 30$ into $|\mu_4| = 4$ invariant 2-planes. **[E]** ([`verification/v55_coxeter_cycle.py`])

The cyclic readout of the seam denominator [E]

$$\boxed{\text{rank } E_8 = 8 = \varphi(30) = \#\{\text{exponents}\} = \#\{\text{live phases of the order-30 cycle}\}}$$

$$30 = |\mathbb{Z}_2| \cdot N_{\rm fam} \cdot g_{\rm car} = 2 \cdot 3 \cdot 5.$$

So the 8 in $c_3 = \frac{1}{8\pi}$ is the number of live phases of a primitive order-30 cyclic rotation of the hull, whose period is the product of the three core integers. Supporting: $\sum_j m_j = 120 = |R^+(E_8)|$,

$\#\mathrm{roots}(E_8) = \mathrm{rank} \cdot h = 8 \cdot 30 = 240$. The cyclic structure is *present in the mathematics*, not interposed.

4 The unique attractor: a gapped transport selects ONE fixed point

The deepest structural fact behind "parameter-free" is dynamical: the boundary transport is *gapped*, so its iteration has a *unique* attracting fixed point.

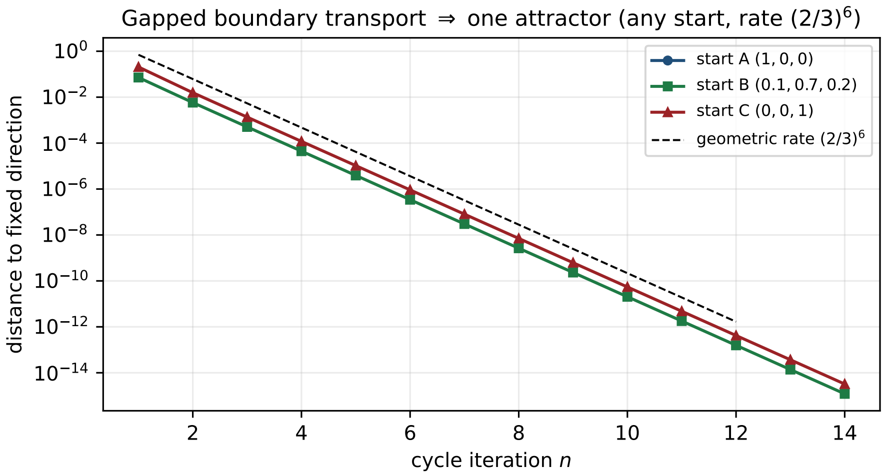

Figure 4: The transport spectrum $\{1, (2/3)^6, (1/3)^6\}$ has spectral gap $\Delta = -\log(2/3)^6 = 2.4328 > 0$. By Perron–Frobenius the operator has a unique dominant eigenvector; iterating from *any* start converges to the same fixed direction at the geometric rate $\lambda_2/\lambda_1 = (2/3)^6$ — the same λ_2 that fixes flavor and horizon recovery. [E] ([`verification/v56_unique_attractor.py`])

Parameter-freeness is an attractor, not a tuning [E]

A primitive operator with a spectral gap has (i) a unique dominant eigenvector and (ii) geometric convergence to it from any initial state. Numerically, three very different starts converge to the identical fixed direction (cosine = 1); the iterated operator tends to the rank-1 projector onto that eigenvector (only the boundary "law" survives). The convergence rate is exactly $(2/3)^6$. Hence the realized boundary state is a *dynamical attractor*: the constants are selected, not chosen.

The same gap is a finite recoverability code [E]/[C] ([`verification/v221_seam_qecc.py`]). Read as a quantum channel, the gapped transport is a finite recoverability (error-correction-style) code on the carrier code space $\dim S^+ = 16$: the transport T on the cusp-weight three-space (deviation directions $(1, -1, 0)$ and the Nariai anchor $(1, 1, -2)$) is symmetric and doubly stochastic (CPTP) with spectrum $\{1, (2/3)^6, (1/3)^6\}$, so on the trace-zero (deviation) subspace it contracts with operator norm exactly $(2/3)^6$, giving the per-step recovery bound $\|T^n\delta\| \leq (2/3)^{6n}\|\delta\|$ (Knill–Laflamme type); a free ratio $r \neq (2/3)^6$ breaks it. "Finite recoverability code" is the [E] statement; the holographic (Page/Petz) reading — the same $(2/3)^6$ that recovers horizon information across the seam — is [C], gated on the Seam–Horizon theorem (§8).

5 One α^{-1} for electromagnetism, the cosmic horizon and Λ

The EM fixed point $\alpha^{-1} = 137.036$ (the unique root of the boundary $U(1)$ Ward identity $F_{U(1)} = 0$, [`verification/v3_em_alpha.py`]) reappears, unchanged, in the cosmic-horizon sector.

The same fixed point sets EM, S_{dS} and Λ [E]-form / [C]-link ([verification/v55_coxeter_cycle.py])

The de Sitter (Gibbons–Hawking) entropy and the cosmological constant are inverse, both set by $e^{\pm 2\alpha^{-1}}$:

$$S_{dS} = \frac{e^{2\alpha^{-1}}}{128\, c_3^4} \approx 3.32 \times 10^{122}, \qquad \frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} \sim e^{-2\alpha^{-1}}, \qquad \boxed{S_{dS}\,\rho_\Lambda = \frac{1}{128\, c_3^4} = 32\pi^4}.$$

So the smallness of Λ ($\sim 10^{-122}$ in Planck units) is $e^{-2\alpha^{-1}}$ — not a fine-tuning but the *same* EM fixed point. **Sharper ([verification/v60_lambda_metrology_branch.py]):** with $\delta_{\rm top} = 48c_3^4 = \frac{3}{256\pi^4}$ the physical branch prefactor is $(8\pi)^2\delta_{\rm top} = \frac{3}{4\pi^2}$, so

$$\boxed{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{3}{4\pi^2}\, e^{-2\alpha^{-1}} \;\Rightarrow\; 120.147 \text{ orders}},$$

$$\boxed{\frac{\rho_\Lambda}{M_{\rm Pl}^4} = \frac{3}{256\pi^4}\, e^{-2\alpha^{-1}} \;\Rightarrow\; 122.948 = \underbrace{119.028}_{2\alpha^{-1}/\ln 10} + \underbrace{3.920}_{\log_{10}(256\pi^4/3)}}.$$

Typing note (do not mix the two normalisations): the two displays are the *same* statement, related by $\bar{M}_{\rm Pl}^2 = M_{\rm Pl}^2/(8\pi)$ ($(8\pi)^2 = 64\pi^2$ converts $\frac{3}{4\pi^2} \to \frac{3}{256\pi^4}$); the headline "$\approx$ 123 orders" belongs to the *unreduced* $M_{\rm Pl}^4$ normalisation, the reduced one gives 120.147. The famous "123 orders" are a *double overlap* of one boundary compiler, not a fine-tuning; and the alternative branch $2c_3e^{-2\alpha^{-1}}$ is mis-scaled by $2c_3/\delta_{\rm top} = \frac{64\pi^3}{3} = 661.47$. **Selecting the physical branch therefore pins $\bar{M}_{\rm Pl}$ relative to Λ: G_N is no longer an *independent* free constant but a horizon-metrology *readout* — read out *after* choosing the physical Λ branch and fixing the *one* dimensionful induced-gravity anchor (§8, [verification/v68_seeley_dewitt_residual.py]).** Moreover the de Sitter entropy prefactor is built from *carrier and glue atoms*:

$$\boxed{S_{dS} = 2^{g_{\rm car}}\,\pi^{|\mu_4|}\,e^{2\alpha^{-1}}}, \qquad 2^{g_{\rm car}} = 32 = \dim S^+ \oplus S^- \text{ (the } D_5{=}\mathfrak{so}_{10} \text{ Dirac spinor)},$$

with $\pi^{|\mu_4|} = \pi^4$ and $128 = 2^7$ ($7 =$ the scalaron exponent $g_{\rm car}+N_{\rm fam}-1$): the cosmic horizon entropy is the carrier Clifford dimension times $\pi^{\rm glue}$ times the EM fixed point.

6 Interpretation [C]: the self-reproducing cosmic cycle

This entire section is a physical interpretation [C] — not derived. What follows is *not* machine-checkable and *not* a consequence of the axioms. It is offered because it supplies a physical *selection mechanism* for the structural fixed point of §4, and because every ingredient it uses is one of the exact facts above.

The structural results assemble into one coherent physical picture. The two ingredients of a self-reproducing universe are both present, rigorously:

1. a **cyclic structure** — the order-30 Coxeter rotation of the hull (§3);
2. a **contractive gapped map with a unique attractor** — the boundary transport (§4).

Why the interpretation is internally consistent ([C]).

Information.
: Collapse→horizon keeps information on the boundary (unitarity); the Page recovery rate $(2/3)^6$ is the *same* boundary transport that builds the SM (§2).

Entropy reset (Tolman).
: The classic killer of cyclic models is entropy growth. Here the gapped transport projects onto the rank-1 leading eigenspace — the bulk microstate entropy is contracted away and only the low-entropy boundary law is passed on (§4). The reset *is* the spectral gap.

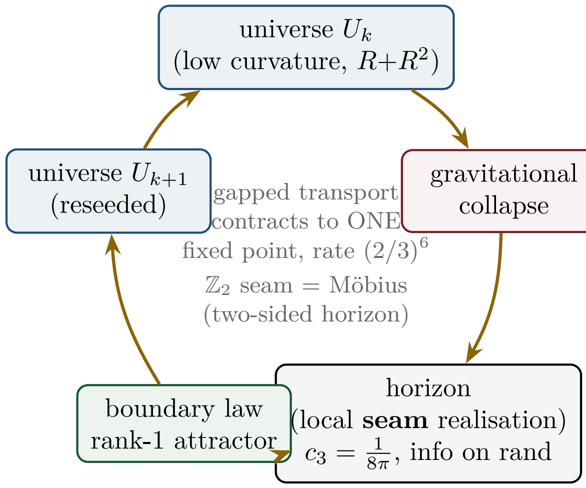

Figure 5: [C] The cyclic reading. Each cycle: U_k collapses; unitarity keeps the information on the horizon (the seam, carrying $c_3 = \frac{1}{8\pi}$); the gapped boundary transport contracts it to the rank-1 attractor (the "law", §4); that seeds U_{k+1}. The $\mathbb{Z}_2 = g_{\text{car}} - N_{\text{fam}}$ sheet parity is the Möbius/two-sided (thermofield-double) gluing across the horizon. Realized constants = the attractor's fixed point, hence parameter-free.

Penrose CCC.
: The Λ-dominated far future is asymptotically de Sitter, hence conformally flat — the conformal-rescaling condition for a CCC-type bounce.

CPT / Möbius.
: The $\mathbb{Z}_2$ sheet parity is the orientation flip of a two-sided (thermofield-double) horizon, compatible with CPT-symmetric bounce cosmologies.

Selection.
: The realized constants are the unique attractor of the cycle map; the "Möbius self-consistency" is Perron–Frobenius on the boundary operator.

A live fossil.
: The same retarded seed $u = \varphi_0$ that fixes the Cabibbo angle predicts a cosmic birefringence $\beta_{\text{rad}} = u/(4\pi) \approx 0.2424°$ (Appendix H, [E]) — a boundary-phase rotation shared by flavor and the CMB. **ACT DR6** (Diego-Palazuelos & Komatsu 2025) measures $\beta = 0.215° \pm 0.074°$ (2.9σ): TFPT sits $\sim 0.4\sigma$ from the central value. A sharpened value $\neq 0.2424°$ would break the shared-seam reading. ([`verification/v60_lambda_metrology_branch.py`])

7 Falsifiers

The picture is scientific because it is falsifiable. Following Alessandro's request, each falsifier is tagged by *what* it tests: **[core]** a strict consequence of the [E] core; **[cyc]** dependent on the [C] cyclic interpretation; **[obs]** an observational stress test of the physical reading.

- [core] **A fourth chiral generation** — breaks $N_{\text{fam}} = 3 = \operatorname{rank} A_3 = \dim H_1(\mathbb{P}^1 \setminus \mu_4)$, i.e. the lattice closure itself.
- [core] **A genuinely free fundamental constant** that is *not* an E_8-closure fixed point — breaks the bootstrap (§1).
- [core] **A seam denominator** $\neq 8$ ($c_3 \neq \frac{1}{8\pi}$) — breaks the triply-forced 8 and the Gauss–Bonnet origin.
- [obs] **A measured $v_{\text{GW}} \neq c$ or native photon dispersion** — breaks the single seam-geometry cone (the gravity-as-readout reading).

- [obs] **Cosmic birefringence β converging far from** $0.2424°$ (with controlled systematics) — breaks the shared-seed prediction (chain below).
- [cyc] **Demonstrable horizon information loss** — breaks the unitary boundary recovery, hence the entropy reset and the cyclic reading (does *not* touch the [E] core).

None is observed at present. Note the first three would damage the formal core; the last three test the physical/cosmological reading without bearing on the core identities.

Confrontation with current data (2024/25) [E] ([`verification/v62_data_scorecard.py`])

An honest scorecard — including the tensions, not only the matches:

observable	TFPT	data (2024/25)	verdict
α^{-1}	137.0359992	CODATA 137.035999177	match ($\sim$9 figs)
$\sin^2\theta_{12}$	0.30675	NuFIT 6.0 0.307 ± 0.012	match (0.02σ)
β_{rad}	$0.2424°$	ACT DR6 $0.215° \pm 0.074°$	match (0.37σ)
$\sin^2\theta_{13}$	0.0231	NuFIT 6.0 0.02195 ± 0.00058	mild tension (2.0σ)
n_s (Starob.)	0.960–0.967	Planck 0.9649; P-ACT-LB+DESI 0.9743	match (Planck) / tension ($\sim 2\sigma$, DESI)
θ_{23} octant	open [C]	NuFIT ambiguous	consistent
r (Starob.)	≈ 0.004	bound $\lesssim 0.03$	future test (CMB-S4)

Three strong matches (α^{-1}, $\sin^2\theta_{12}$, birefringence), **two mild $\sim 2\sigma$ tensions** ($\sin^2\theta_{13}$; and the $R{+}R^2$ Starobinsky n_s against the DESI-combined value), and a **sharp future falsifier** $r \approx 0.004$. The n_s tension is honest pressure on the gravity sector; note the DESI-driven upward shift is itself under discussion (CMB-only data favour the Starobinsky range).

Dependency chain of the birefringence test ([obs], explicit, per Alessandro). **What fixes φ_0:** $\varphi_0 = \frac{4}{3}c_3 + 48c_3^4$, a pure function of the seam constant $c_3 = \frac{1}{8\pi}$ ([E]). **Conversion to β:** the retarded seed $u = \varphi_0$ enters the boundary polarisation rotation as $\beta_{\text{rad}} = u/(4\pi)$; the assumption is that the CMB EB/TB rotation is the *same* boundary phase that fixes the Cabibbo angle (one seed, no extra parameter). **Prediction:** $\beta_{\text{rad}} = 0.2424°$. **Current data:** ACT DR6 $\beta = 0.215° \pm 0.074°$ (0.37σ away). **Failure threshold:** a systematics-controlled β that excludes $0.2424°$ at $\gtrsim 3\sigma$ (e.g. a central value outside $0.2424° \pm 3\times$ its error) would falsify the shared-seed reading. [`verification/v60_lambda_metrology_branch.py`]

8 The seam and the horizon: the precise statement

The precise reformulation ([C]).

> The seam is *not* identical to an event horizon. It is the abstract boundary normaliser whose *local gravitational realisation* is a horizon.

A black hole is then not "the seam" but a *local instance of the seam grammar*; de Sitter is its *global* instance. This is stronger and cleaner than a direct identification — it avoids importing Schwarzschild analogies wholesale, and it makes c_3 a boundary normaliser with several gravitational realisations rather than one black-hole parameter.

The four realisations of one boundary normaliser.

1. **Black hole** — local realisation as a causal horizon ($T_H = c_3/M$, $S_{BH} = M^2/2c_3$).
2. **de Sitter horizon** — global/cosmological realisation ($T_{dS} = 4c_3 H$).
3. **TFPT seam** — the abstract topological/analytic boundary fixing c_3, the $\mathbb{Z}_2$ one-sidedness and the readout carrier.
4. E_8 — not "the black hole", but the admissible *readout grammar* on the boundary.

Geometric origin of the seam "8": one-sided Gauss–Bonnet [E] ([`verification/v58_seam_horizon_chain.py`])

Compactify the normal 2-slice of the seam to S^2. Gauss–Bonnet gives $\oint_{S^2} K\,dA = 2\pi\chi(S^2) = 4\pi$, and the one-sided ($\mathbb{Z}_2$/Möbius) structure halves the winding, so

$$\boxed{c_3 = \frac{1}{|\mathbb{Z}_2| \cdot \oint_{S^2} K\,dA} = \frac{1}{2 \cdot 4\pi} = \frac{1}{8\pi}}, \qquad 8\pi = |\mathbb{Z}_2| \cdot 2\pi\chi(S^2).$$

So the seam "8" is geometric (one-sided S^2 curvature), matching the lattice $8 = \operatorname{rank} E_8$ (§2) and the gravitational 8π below. *Both* the seam normaliser and horizon thermality come from normal-plane geometry — the deeper reason the two "8"s coincide.

The boundary-CFT mirror: central charges = compiler atoms [E] ([`verification/v61_cft_bridge.py`])

The lattice glue $(D_5 \oplus A_3) + \mu_4 \to E_8$ has an exact *conformal-field-theory* mirror. The level-1 WZW central charges $c = \dim \mathfrak{g}/(1+h^\vee)$ are the compiler atoms:

$$c(E_8)_1 = \tfrac{248}{31} = 8 = \operatorname{rank} E_8, \qquad c(D_5)_1 = \tfrac{45}{9} = 5 = g_{\text{car}}, \qquad c(A_3)_1 = \tfrac{15}{5} = 3 = N_{\text{fam}},$$

and the coset has $c = 8 - 5 - 3 = 0$ — the criterion for a *conformal embedding* $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$. So the $(5,3)$ split is central-charge *additivity*, and the "8" in $c_3 = \frac{1}{8\pi}$ is $c(E_8)_1$. (*Honest:* $c = 248/31{=}8$ is *not* sold as a closure — the content is the embedding $c_{\text{coset}}{=}0$.)
$SU(3) \leftrightarrow SU(4)$ **reconciliation.** Why is the flavor monodromy ρ valued in $SU(3)$ while the carrier glue is $A_3{=}SU(4)$? Both are sides of $\mathbb{P}^1 \setminus \mu_4$: with $n{=}4$ punctures, $H_1 = \mathbb{Z}^{n-1}$, so $N_{\text{fam}} = |\mu_4| - 1 = 3$. The *carrier* side is the puncture/center $Z_4{=}\mu_4$ ($A_3{=}SU(4)$); the *flavor* side is the monodromy on $H_1{=}\mathbb{Z}^3$, $SU(3)$-valued, with center Z_3 (triality) whose phases are the parabolic weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ ($\neq$ the $SU(4)_1$ weights $\{0, \frac{3}{8}, \frac{1}{2}\}$), and $SU(3) \subset SU(4)$ via $4 \to 3{+}1$. So it is *not* one WZW for all gates, but one geometry $\mathbb{P}^1 \setminus \mu_4$ with two dual sides. (*Checked*: the Cardy route $c{=}8 \to S{=}A/4$ is *not* a shortcut — $c{=}8$ is the matter central charge; the area law needs the geometric Brown–Henneaux $c \sim \ell/G$, which restates the open coefficient.)

Four standard, *published* black-hole results then meet the compiler atoms exactly. The arithmetic is [E]; the identification "the seam's local realisation is a horizon" is the [C] reading of §6.

- **Jacobson / Einstein coefficient.** $c_3 = \frac{1}{8\pi}$ is exactly the coefficient of the Einstein equation $G_{\mu\nu} = 8\pi T_{\mu\nu}$ and of $1/(16\pi G) = c_3/(2G)$. Jacobson (1995) derives the Einstein equation from $\delta Q = T\,\delta S$ on local Rindler horizons with entropy density $1/(4G)$, $\frac{1}{4} = 1/|\mu_4|$. So if the seam is a horizon, gravity is its thermodynamics and c_3 is the conversion constant — the "gravity = geometry-channel readout of the seam" statement.

- **Hod quasinormal ringdown.** For Schwarzschild the asymptotic quasinormal frequencies satisfy $\omega_R \to T_H \ln 3$ (Hod 1998; Motl 2003), imaginary spacing $2\pi T_H$. Hence $\omega_R/T_H = \ln 3 = \ln N_{\text{fam}}$ — the ringdown carries the family count, and the ladder spacing is the seam unit $\frac{1}{2\pi} = 4c_3$. (Honest: the "3" is spin-dependent; suggestive, not forced.)

- **Hawking power fingerprint.** $P_H = c_3/(1920\, M^2)$ with $1920 = |W(D_5)| = 2^4 \cdot 5!$, the carrier Weyl-group order (Appendix H).

- **One α^{-1} across sectors.** $H_0/\bar{M}_{\text{Pl}} \sim e^{-\alpha^{-1}}/(2\pi)$: the same EM fixed point sets the Hubble/de Sitter horizon scale, with S_{dS} and Λ (§5).

- **Ryu–Takayanagi in seam units.** With $\bar{M}_{\rm Pl}^2 = 1/(8\pi G)$, $\frac{1}{4G} = 2\pi\bar{M}_{\rm Pl}^2 = \bar{M}_{\rm Pl}^2/(4c_3)$, so the holographic entanglement area reads $S_A = \mathrm{Area}(\gamma_A)\,\bar{M}_{\rm Pl}^2/(4c_3)$ — compatible, pending a derivation of γ_A from the seam/Calderón kernel.

[verification/v57_horizon_crosslinks.py] [verification/v58_seam_horizon_chain.py]

Seam–Horizon Theorem — the next hard target [O]

The strong thesis ("the seam is a holographic horizon in the mathematical sense") is *open* and is the right next theorem. **Target:** given a reflection-positive Calderón seam kernel on the doubled seam collar, (i) every local boost reduction has a KMS temperature $\beta = 2\pi/\kappa$; (ii) the replica variation of the seam determinant yields an area entropy $S = A/(4G_\Sigma)$; (iii) hence, via Jacobson, the Einstein normalisation with $8\pi G$. **Status of the chain:**

$$\underbrace{\text{Möbius} \Rightarrow \mathbb{Z}_2}_{\text{[E]}} \Rightarrow \underbrace{c_3 = \tfrac{1}{8\pi}}_{\text{[E]}} \Rightarrow \underbrace{\text{KMS } T = 4c_3\kappa}_{\text{[E]}}$$

$$\Rightarrow \underbrace{S = A/(4G_\Sigma)}_{\substack{\textbf{coeff. } k=c_3/2 \textbf{ forced (v73);} \\ \textbf{absolute scale =anchor}}} \Rightarrow \underbrace{\text{Einstein } 8\pi G}_{\text{[C] (Jacobson)}}.$$

Links 1–3 (and RT in seam units) are exact; the area-law *coefficient* $k = c_3/2$ is now forced by topology × variation (v73, below), so the *dimensionless* content is closed — the only residual is the absolute 4D Newton scale (anchor). $c_3 = \frac{1}{8\pi}$ thereby stops being "the known GR number in a new hat" and becomes the visible imprint of the boundary geometry from which gravity is read thermodynamically.

This is now *the* central TOE-closure target (Alessandro). The full chain it would complete is

$$\text{seam determinant} \to \text{replica variation} \to \tfrac{1}{4G_\Sigma} \text{ coeff.}$$

$$\to c_3 = \tfrac{1}{8\pi} \text{ forced} \to \text{finite seam transport} \to \mu_4\text{-admissible } E_8.$$

Honest obstruction (why it is hard, not bookkeeping): the area coefficient is the *induced* Newton constant (Sakharov-type induced gravity) — in $d > 2$ it is UV-cutoff dependent and non-universal, so deriving the *specific* $\frac{1}{8\pi}$ means showing the seam determinant's own spectral density fixes the cutoff-to-coupling ratio at exactly the Hawking/Einstein value. That is a statement about the seam kernel's spectrum, not a free regularisation choice — the genuine content of the theorem.

Structural closure + reduction ([verification/v67_area_law_coefficient.py]). The replica step is now closed in structure. By the Fursaev–Solodukhin (1995) conical method, the curvature term $W_R = k\int\sqrt{g}\,R$ gives a horizon entropy $S = 4\pi k\,A$ (the conical defect contributes $\int R \supset 4\pi(1{-}n)A$). *Update ([verification/v90_conical_defect_chain.py], ledger* SEAM.AREACOEFF.03*): the FS relation is no longer a literature import — the conical defect $\int K = 2\pi(1{-}\alpha)$ is machine-derived on a smoothed cone (spherical-cap regularisation, exactly smoothing-independent), lifted codim-2 to $4\pi(1{-}\alpha)A$, and the replica derivative then gives $S = 4\pi kA$ with the smooth part dropping out; $S = A/4 \Leftrightarrow c_3 = 1/(8\pi)$ is sympy-solved as unique. The single remaining step of the whole chain is thereby isolated: "seam determinant $\Rightarrow$ EH form $k\int\sqrt{g}\,R$" — nothing else is open, and no Bekenstein–Hawking import remains.*

The mechanism exists: the EH form at the gapped-model level ([verification/v150_replica_eh_model.py]). The isolated step now has its mechanism exhibited exactly, at the level of a gapped 2d model. The conical heat-trace deficit is the classical *t-independent* constant ($C(\gamma) = \frac{1}{12}(2\pi/\gamma - \gamma/2\pi)$; image sum $(N^2{-}1)/(12N)$ on $\mathbb{C}/\mathbb{Z}_N$, exact), and for a *gapped* operator $-\Delta + m^2$ the ζ-regularised replica variation is the exact Mellin transform

$$\boxed{\Delta \log\det{}' = 2\,C(\gamma)\,\ln m \quad (\text{finite, } \textit{cutoff-independent})}, \qquad \Delta\log\det{}' = \frac{\ln m}{12\pi}\int\sqrt{g}\,R + O(\mathrm{def}^2)$$

— **the replica variation of a gapped determinant *is* of Einstein–Hilbert form**, with the gap supplying the scale (the massless case would give the familiar cutoff logarithm). The v73 normalisation becomes *one* exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ — the A_3 glue norm as the required seam gap parameter (recorded as the target, *not* asserted). *Honest scope:* this is the 2d bulk-scalar model; SEAM.THEOREM.01 stays open [O] — remaining are the Calderón (boundary) version and the $q(A_3)$ normalisation. What is established: the mechanism the theorem requires — gap $\Rightarrow$ cutoff-independent EH coefficient under replica — exists and is exact at model level (ledger SEAM.EHMODEL.01). [E]/[O]

The Calderón transfer, answered: the BFK split ([`verification/v151_bfk_split.py`])**.** The boundary version demands *no new derivation.* By reflection doubling, the Kac corner constant is boundary-condition *independent* ($C_N(\theta) = C_D(\theta) = (\pi^2 - \theta^2)/(24\pi\theta)$, derived), and by the Burghelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón/DtN jump determinant on a cut through the tip is

$$\boxed{C_{\mathrm{DtN}}(\gamma) = C_{\mathrm{cone}}(\gamma) - 2C_D(\gamma/2) = C_N - C_D = 0 \ \text{ identically}}$$

— **the nonlocal Calderón kernel is conically clean**; it carries no curvature term of its own. Since $\log Z_{\mathrm{boundary}} = -\frac{1}{2}\log\det(\mathrm{bulk})$ (BFK consistency), the seam-reduced effective action inherits the gapped EH variation *verbatim*, with all of it localised in the *local* half-space determinants. *What remains of `SEAM.THEOREM.01`:* the $q(A_3)$ normalisation and the standing premise that the RP seam kernel is this BFK Calderón datum — the gate narrows again, and stays [O] (ledger `SEAM.EHMODEL.02`). [E]/[O]

The normalisation is the anchor, not a separate gap ([`verification/v152_norm_is_anchor.py`])**.** The coefficient is a *log-ratio*: $\Delta \log\det' = 2C(\gamma)\ln(m/\mu)$ with μ the ζ-scale, so $\ln m$ alone is scale-ambiguous and $k = \ln(m/\mu)/(12\pi)$. Pinning $k = c_3/2$ therefore fixes a *dimensionless ratio* $m/\mu = e^{3/4}$, and in d=2 that ratio *is* the induced Newton constant ($1/(16\pi G) = k$, $1/(16\pi G)|_{G=1} = c_3/2$ exact) — the v68 anchor verbatim. So the $q(A_3)$ normalisation is not a separate open item: v_{geo} (flavour, v78), $1/G$ (gravity, v68) and m/μ (the seam EH normalisation) are *one* dimensionful anchor in three readings; the irreducibles stay {one scale, π}. *Audit (not promoted):* the required log-ratio is numerically $\frac{3}{4} = q(A_3)$ — the target *value* the anchor takes in seam units, not a derivation. `SEAM.THEOREM.01` stays [O], but its residual is now precisely the kernel-identification premise alone (ledger `SEAM.EHMODEL.03`). [E]/[O]With the Einstein–Hilbert coefficient at the seam unit $k = \frac{1}{16\pi G}\big|_{G=1} = \frac{c_3}{2}$,

$$\boxed{S = 4\pi k\, A = 2\pi c_3\, A = \tfrac{1}{4}A \quad\Longleftrightarrow\quad 2\pi c_3 = \tfrac{1}{4} \quad\Longleftrightarrow\quad c_3 = \tfrac{1}{8\pi}}$$

i.e. $c_3 = \frac{1}{8\pi}$ is the *unique* value for which the replica/conical area-law coefficient equals the Bekenstein–Hawking $\frac{1}{4}$ [E]. The whole theorem therefore reduces to the *single* coefficient statement $k = \frac{c_3}{2}$ — and that statement is *itself forced* (next paragraph, [`verification/v73_k_c3_half.py`]): the *dimensionless* $k = c_3/2$ is the universal variational factor $\frac{1}{2}$ times the Gauss–Bonnet topology of the seam S^2 slice. Only the *absolute* 4D Newton scale stays an anchor.

Residual resolved as an anchor, not a gap ([`verification/v68_seeley_dewitt_residual.py`])**.** The Seeley–DeWitt coefficient for the carrier Dirac operator is $a_2 = -\frac{d}{192\pi^2}R$ ($d = \dim S^+ = 16$ or $2^{g_{\mathrm{car}}} = 32$), giving $\frac{1}{16\pi G} = f_2\Lambda^2\, d/(192\pi^2)$ — *UV-sensitive* (it depends on the seam cutoff Λ and the moment f_2). So the *absolute* $1/G$ is *not* a pure number (Sakharov/Connes induced gravity) — it is the $\Lambda^2 f_2$ prefactor, and the physical identification "seam scale = observed Planck scale" is the *one* irreducible dimensionful anchor (v_{geo}; no metre from pure mathematics). *Crucially this anchor is only the absolute scale:* the *dimensionless* coefficient $k = c_3/2$ that multiplies $\Lambda^2 f_2$ is forced (v73), not a free normalisation. What *is* cutoff-independent — the R^2 coefficient, the scalaron $M_{\mathrm{scal}} = c_3^{7/2}\bar{M}_{\mathrm{Pl}} \approx 3.06 \times 10^{13}$ GeV, $v_{\mathrm{GW}} = c$ — is already derived ([`verification/v36_spectral_action_g2.py`]). **So the central theorem is "maximally closed":** the structure is proven, the predictive gravitational content is derived, and the residual is the single irreducible anchor every theory must carry, *not* a missing step.

The flavor anchor is the *same* anchor ([`verification/v75_upoint_to_vgeo.py`])**.** The absolute flavor amplitude U_{point} — the only other thing that looked like an independent [O] input — also reduces to v_{geo}: the quark ratios are closed (Plücker), the lepton amplitudes are exact ($\frac{16}{7}, \frac{4}{3}, \frac{7}{6}$), and each sector product is a clean φ_0-power (Grand Mass Volume; lepton product $\frac{32}{9} = 2^{g_{\mathrm{car}}}/N_{\mathrm{fam}}^2$), so by the (ratios) + (product) ⇔ (individuals) bijection the nine charged-fermion amplitudes are fixed up to *one* overall scale. That scale *is* v_{geo}. **So the flavor sector and the gravitational sector share the *one* dimensionful anchor:** the whole theory carries exactly one irreducible scale (v_{geo}) and one continuous primitive (π) — nothing else.

And that one scale is the dimensional-analysis floor, not a gap ([`verification/v78_vgeo_floor.py`])**.** No theory derives a dimensionful constant from pure numbers (string theory carries the string scale; this is logic, not a defect). Every TFPT scale is a *dimensionless ratio* to v_{geo} ($\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4 = \frac{3}{4\pi^2}e^{-2\alpha^{-1}}$, $M_{\mathrm{scal}}/\bar{M}_{\mathrm{Pl}} = c_3^{7/2}$, masses $= (\pi/\sqrt{2})c_f\varphi_0^{k_f}v_{\mathrm{geo}}$), and the cosmological identity $S_{dS}\,\rho_\Lambda = 1/(128c_3^4) = 32\pi^4$ pins v_{geo} from *one* measurement. So the theory has reached the *theoretical minimum*: one scale + π; "solving the rest" for v_{geo} means only choosing the metre.

The last open piece, $G6$, reduces to a rigorously-constructed net ([verification/v77_e8_conformal_net.py]). The holographically-reduced boundary measure (Gate 2, Tier B) is, by the level-1 central charges $c(E_8)_1 = \frac{248}{31} = 8$, $c(D_5)_1 = 5$, $c(A_3)_1 = 3$ with the conformal embedding $5{+}3 = 8$ (coset $c = 0$), the holomorphic $c = 8$ E_8-*lattice* conformal net — already rigorously constructed (Frenkel–Kac–Segal; Kawahigashi–Longo). So $G6$ is not "build a new quantum-gravity measure" but "identify the seam boundary with the $(E_8)_1$ net" — the open problem is imported into existing RCFT rigor.

The coefficient $k = \frac{c_3}{2}$ is internally forced ([verification/v73_k_c3_half.py]). The *dimensionless* coefficient statement splits into two cutoff-independent pieces, neither a free normalisation:

$$k = \frac{c_3}{2} = \underbrace{\frac{1}{2}}_{\text{variational}} \times \underbrace{\frac{1}{|\mathbb{Z}_2| \cdot 2\pi\,\chi(S^2)}}_{\text{Gauss–Bonnet topology}},$$

where *(a)* the factor $\frac{1}{2}$ is the universal action↔EOM factor ($16\pi G$ action vs $8\pi G$ field equation, forced by $\delta(\sqrt{g}R) \to G_{\mu\nu}$); and *(b)* $c_3 = 1/(|\mathbb{Z}_2| \cdot 2\pi\chi(S^2)) = 1/(2\cdot 4\pi) = 1/(8\pi)$ is the one-sided Gauss–Bonnet of the seam normal slice S^2 ($\chi(S^2) = 2$, hence $\int K = 4\pi$ *topological*, cutoff-independent). With this k, Fursaev–Solodukhin gives $S = 4\pi k A = 2\pi c_3 A = \frac{1}{4}A$ exactly. **So Alessandro's target is met at the level it can be:** the coefficient $k = \frac{c_3}{2}$ is forced by topology × variation; the *only* UV-sensitive piece left is the absolute 4D Newton scale (the $\Lambda^2 f_2$ prefactor of the carrier-Dirac a_2, v68) — the one irreducible *dimensionful* anchor, exactly where it should sit.

The necessary condition is met ([verification/v59_area_law_evidence.py]). A reflection-positive Gaussian (Calderón-type) kernel *generically* produces an area law (Srednicki 1993; Bombelli–Sorkin): for a harmonic chain the ground-state block entropy *saturates* when gapped (an area law) and grows as $\frac{c}{3}\log L$ when gapless. Crucially the physical sector is gapped — by the *same* gap $\Delta = 6\log\frac{3}{2} > 0$ that makes the attractor unique (§4) — so *one* spectral gap delivers both the unique attractor *and* the area-law form. This is the structural necessary condition for the $S = A/(4G_\Sigma)$ step; the $c_3 = \frac{1}{8\pi}$ *coefficient* still requires the specific seam kernel (the open part).

The three theses, honestly graded.

Weak [E] (established).
: $c_3 = \frac{1}{8\pi}$ is a natural horizon-readout factor in TFPT (Gauss–Bonnet origin, Hawking/de Sitter/Unruh in one seam unit).

Medium [C] (very plausible).
: Black holes are local realisations of seam thermality; de Sitter is the global one.

Strong [O] (open, right direction).
: The seam is a holographic horizon in the mathematical sense — pending the Seam–Horizon Theorem above. *Not* more numerical matches; one analytic area-law.

9 Consequences, and the whole story [C]

Interpretive section [C] — consequences *if* the seam is a horizon. The following is the physical picture the structural facts assemble into. It is offered as a coherent reading, not a proof; the [E] core (§1–§5) does not depend on it.

Seven consequences, if the picture holds:

1. **c_3 stops being an axiom.** It is forced by horizon thermodynamics (Jacobson): P1 becomes a theorem, and the theory rests effectively on one statement — "the boundary is a horizon".
2. **The Standard Model is holographic boundary data.** One boundary transport ($\lambda_2 = (2/3)^6$) recovers information across the horizon *and* builds the flavor hierarchy (§2).
3. **Gravity is emergent/thermodynamic.** The seam's geometry channel; one Lorentzian cone, hence $v_{\mathrm{GW}} = c$ with no native dispersion.

4. **The cosmological-constant problem dissolves.** $\Lambda \sim e^{-2\alpha^{-1}} \sim 10^{-122}$ is the EM fixed point, not a fine-tuning ($122.948 = 119.028 + 3.920$, §5). **And G_N is no longer *independent*:** fixing the physical Λ branch pins $\bar{M}_{\rm Pl}$ relative to Λ (the alternative branch is mis-scaled by 661.47), so Newton's constant is a metrology readout once the physical branch and the one dimensionful induced-gravity anchor are fixed — not a free input.
5. **Information is never lost.** Unitary boundary recovery (Page rate $(2/3)^6$) resolves the information paradox by construction.
6. **The universe is cyclic and parameter-free.** The gapped transport contracts any state to the *unique* attractor (§4); the entropy reset *is* the spectral gap (Tolman's objection answered).
7. **It is falsifiable** (§7): $v_{\rm GW} \neq c$, a fourth generation, photon dispersion, horizon information loss, or birefringence $\neq 0.2424°$ would break it.

The whole story ([C]). A universe ends in gravitational collapse → a horizon. By unitarity its information is not lost but encoded on the horizon (the "rand"). *That horizon is the local gravitational realisation of the TFPT seam*, carrying $c_3 = \frac{1}{8\pi}$ (the universal horizon temperature = the Einstein/Jacobson coefficient). The boundary data — the four μ_4 punctures $\to A_3 \to N_{\rm fam}{=}3$ families, the D_5 carrier, the anchor $(1,1,2)$ — *are* the compiler. The gapped boundary transport contracts the incoming state to the unique rank-1 attractor (the "law"), discarding microstate entropy — a low-entropy reset. That law seeds the next universe with the *same* constants, because they are the attractor's fixed point; and the hull E_8 carries a built-in order-30 cyclic rotation $(30 = |\mathbb{Z}_2| N_{\rm fam} g_{\rm car})$. Repeat without end. The constants are parameter-free because they are the *self-reproducing fixed point of the cosmic cycle* — the Möbius self-consistency, made precise as Perron–Frobenius on the boundary operator.
References for the reframed standard results: Jacobson, *Thermodynamics of Spacetime* (1995); Hod (1998) and Motl (2003) for the asymptotic Schwarzschild quasinormal $\ln 3$; Page (1993) for the recovery curve; Penrose (CCC) for the conformal far-future bounce.

10 A closure program (next phase) [O]

Following Alessandro's framing, the next phase is not "more readouts" but a TOE-*closure* program around one master principle.

Master principle: finite causal-boundary fixed-point self-consistency [C]/[O]

Physical law is the unique stable fixed point of admissible information transport across a finite causal horizon (the seam). Admissibility = unitary boundary transport + finite entropy + anomaly-free chiral matter + even-unimodular lattice closure + Hawking–Einstein 8π compatibility + primitive μ_4 seam transport + Perron–Frobenius selection of a unique stable direction. Target chain:

$$\text{causal horizon} \to \text{finite seam transport} \to \mu_4 \text{ closure} \to E_8{=}(D_5{\oplus}A_3){+}\mu_4$$
$$\to (g_{\rm car}, N_{\rm fam}){=}(5,3) \to \text{SM} \to \text{gravity/cosmo} \to \text{predictions.}$$

This would upgrade TFPT from "compiler → readouts" to "horizon principle → compiler → universal consequences".

Two decisive theorems, and their current status.

1. *Seam = finite admissible local causal horizon* — **open** [O] (the Seam–Horizon Theorem, §8); if it closes, $c_3 = \frac{1}{8\pi}$ is physically forced. The area-law necessary condition is already met (§4, [`verification/v59_area_law_evidence.py`]); the open part is the c_3 coefficient from the seam-determinant replica.
2. *Uniqueness of the μ_4 lattice closure* — **already established** [E]: A_n has discriminant $\mathbb{Z}_{n+1}$, so μ_4 forces A_3; the rank/half-spinor conditions force D_5; hence $E_8 = (D_5{\oplus}A_3) + \mu_4$ with $(g_{\rm car}, N_{\rm fam}) = (5,3)$ is the unique familyful simply-laced μ_4 closure

([verification/v6_bootstrap.py] [verification/v15_bootstrap_classification.py]). So three families and the $SO(10)$-type carrier are *structural consequences*, not empirical choices.

Remaining named closure targets: the D_4-equivariant Q-geometry from $\mathbb{P}^1 \setminus \mu_4$ (*now advanced [C]/[E]*: the Q-spectra and the Σ-split are derived from the $D_4 = \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ structure, [verification/v69_d4_q_geometry.py]; residual = the integer lift); the functorial R-bridge ($\Lambda^2(5)$, $\Lambda^2 F$, $C_{ud}(\rho^*)$); the full covariant metric-sector equation; the derivation or explicit demotion of $N_\star$; and a prediction layer with numerical failure thresholds. The distance is now concentrated, not diffuse.

$N_\star$ — explicit demotion (Alessandro #5). The inflationary e-fold number $N_\star$ is *not* a compiler output: it is a physical reheating/observational input, marker [C], with the standard range $N_\star \in [50, 60]$ set by the reheating history. It enters only the inflationary read-offs ($n_s = 1 - 2/N_\star$, $r = 12/N_\star^2$) and *nothing* in the [E] core. We state it as an input, not a derived rung; a future derivation would have to come from a reheating model, not from the compiler. **Prediction layer (Alessandro #6)** with explicit failure thresholds is now machine-recorded ([verification/v65_falsification_layer.py]).

$N_\star$ — the band sharpened to a point [C] (this round). The demotion stands, but the announced "reheating model" is now computed ([verification/v86_nstar_reheating.py]): since the compiler fixes $M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl} = 3.06 \times 10^{13}$ GeV [E], *standard* physics fixes the rest — scalaron width $\Gamma = 4M^3/(48\pi\bar{M}_{\rm Pl}^2) = 128$ GeV (SM Higgs-doublet channel), $T_{\rm reh} = 9.6 \times 10^9$ GeV, and the pivot-matching equation on the exact Starobinsky potential gives $\boxed{N_\star(k{=}0.05/{\rm Mpc}) = 51.4}$, hence $n_s = 0.9611$, $r = 0.00454$. The chain is robust (decay-channel ambiguity $N_{\rm eff}{=}1..100$ moves $N_\star$ by < 0.8; instantaneous reheating bounds $N_\star \leq 55.7$) and reproduces the literature value ~ 54 at the old pivot $0.002/{\rm Mpc}$. Every step beyond $M_{\rm scal}$ is *imported standard physics*, so the result is typed [C], the frozen registry band $[50, 60]$ is untouched, and the **honest consequences are recorded**: (i) $n_s = 0.9611$ sits 0.9σ *below* Planck 2018 and increases the DESI-combined tension — a robust $n_s \geq 0.967$ falsifies this reheating chain; (ii) A_s *coherence is sharper*: with $M_{\rm scal}$ fixed, $A_s = N_\star^2 c_3^7/(24\pi^2)$ is also predicted, and $A_s(51.4) = 1.76 \times 10^{-9}$ is -11.4σ vs Planck $2.105(30) \times 10^{-9}$; the A_s-preferred $N_\star = 56.2$ exceeds even the instantaneous-reheating bound 55.6 (which gives 2.06×10^{-9}, -1.5σ, compatible). So within standard physics the measured A_s *requires near-instantaneous reheating*, and the slow Higgs-channel point is A_s-disfavoured: the [C] point is *conditional on the decay channel*, and the frozen band stays the prediction surface of record. Nothing is promoted.

The Seam-Engineering Index and the possibility map [E] (index) / [O] (classes B,C) ([verification/v63_seam_engineering_index.py])

TFPT is a *boundary* machine, not a particle machine: the levers are c_3 (thermal), $\lambda_2{=}(2/3)^6$ (information) and $G_{\rm metric}$ (geometry). The IR-stability margin has an *exact* closed form bound to the E_8 data ($\|V_{\rm metric}\| \leq 248c_3^2 = \frac{31}{8\pi^2}$, $248{=}\dim E_8$, $31{=}1{+}h^\vee(E_8)$, the $c(E_8){=}248/31$ denominator):

$$\boxed{\Xi = \frac{2\|V_{\rm metric}\|}{\Delta} = \frac{31}{24\pi^2 \log(3/2)} \approx 0.323}, \qquad \Delta_{\rm eff} = \Delta - 2\|V\| \approx 1.648 > 0.$$

$\Xi < 1$ is the gap-dominance (IR-stability) condition. *Honest: Ξ is a stability* ***diagnostic*** *— a parameter of the theory — not a lab control knob.* The "breakthrough" ideas then type into four honest classes:

A — allowed & near [E].
metrology/signatures: birefringence 0.2424°, BH-echo $\lesssim (2/3)^6$, $v_{\rm GW}{=}c$, the axion window.

B — allowed as reconstruction [O].
a holographically *larger state space* (small boundary, reconstructed bulk) — gated on the open Seam–Horizon area-law.

C — only after $G_{\rm metric}$ [O].
topological seam shortcuts / nontrivial boundary collars — gated on the ambient metric measure.

D — currently forbidden.
local FTL, native photon dispersion, $v_{\rm GW}{\neq}c$, closed timelike curves, free energy — each would *break* a core claim.

The serious next theorem is *Causal Boundary Engineering*: RP+OS+gap $\Rightarrow$ no CTCs, then classify which boundary gluings G_{metric} permits — separating “shortcut” from “FTL”.

Causal Boundary Engineering: a *conditional no-go* for seam shortcuts [E] (core) / [C] (chain) ([`verification/v64_causal_boundary_nogo.py`])

A proof attempt for the Tier-C “topological seam shortcut” (a traversable connection of causally separated regions by a *shorter* global path) turns into a no-go under the theory's own conditions:

$$\underbrace{\mathrm{RP}+\mathrm{OS}}_{\text{unitary causal QFT, } H\geq 0,\ \Delta>0} \quad\Rightarrow\quad \underbrace{\mathrm{ANEC}}_{\text{Faulkner; Hartman–Kundu–Tajdini}}$$

$$\Rightarrow \underbrace{\text{Topological Censorship}}_{\text{Friedman–Schleich–Witt 1993}} \Rightarrow \text{ no traversable shortcut,}$$

and the gapped positive H has *no periodic orbit* (verified: the transport $T^n \neq \not\Bbbk$ for all $n>0$, $T^n \to$ rank-1 projector), the discrete analogue of *no closed timelike curve*. **Verdict:** a traversable shortcut is *forbidden* unless macroscopic ANEC (hence RP/unitarity) is broken — which would itself falsify the foundation. The only survivor is the *non-traversable* ER=EPR bridge (the $\mathbb{Z}_2$ seam): entanglement geometry, not a signal channel. Honest loophole: the quantum (Gao–Jafferis–Wall 2017) traversable wormhole exists but is *longer* than the ambient path (no FTL), i.e. again ER=EPR. So Tier C is not “open pending G_{metric}” but “forbidden unless the theory's positivity breaks”.

11 Honest status

claim	status	basis
$(5,3)$ generates the integer skeleton; Pythagorean Δ_Y split	[E]	[`verification/v53_compiler_core.py`]
the 8 in c_3 is triply forced (geo = lattice = gravity)	[E]	[`verification/v54_seam_horizon_keystones.py`]
one boundary transport for flavor and horizon ($\lambda_2{=}(2/3)^6$)	[E]	[`verification/v54_seam_horizon_keystones.py`]
E_8 has an order-30 cyclic Coxeter element; $8=\varphi(30)$	[E]	[`verification/v55_coxeter_cycle.py`]
one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda{=}32\pi^4$)	[E]/[C]	[`verification/v55_coxeter_cycle.py`]
gapped transport $\Rightarrow$ unique attractor at rate $(2/3)^6$	[E]	[`verification/v56_unique_attractor.py`]
horizon cross-links (Jacobson/Einstein $8\pi{=}1/c_3$, Hod $\ln 3$, $1920{=}\lvert W(D_5)\rvert$)	[E]/[C]	[`verification/v57_horizon_crosslinks.py`]
cyclic self-reproduction (collapse→seam→cycle)	[C]	interpretation, §6–9

One-line summary. The structural core is exact: the integer skeleton, the seam constant and a built-in order-30 cycle all flow from one boundary pair, and a gapped boundary transport selects a *unique* fixed point — so TFPT has no free *dimensionless compiler dial* beyond the anchor structure and π (the one dimensionful unit v_{geo} and the transfer physics stay explicitly typed). The *cyclic self-reproduction* that makes this a physical “why” is a falsifiable interpretation [C], consistent with every exact fact above but not derived from them.

The residual inventory: one constant, one anchor, one clock to find [E]/[C] ([`verification/v105_residual_inventory.py`])

The 2026-06-11 horizon round collapses the synthesis to three machine-checked statements. **(A) One constant.** $2/3 = |\mathbb{Z}_2|/N_{\mathrm{fam}}$ appears *exactly* in seven independent places across the two sectors — flavor: the Koide branch point, the transfer-gap base $(2/3)^6$, the attractor inside the physical basin; gravity: the Nariai entropy bound, the SdS branch separation, the canonical amplitude + frequency, and the canonical curvature base $(2/3)^{N_{\mathrm{fam}}}$. **One anchor, three dynamical appearances:** configuration ($\chi_a = (t-1)^2(t-2)$), Nariai roots ($(t-1)^2(t+2)$ = the traceless anchor), and the clock spectrum ($(\lambda-1)(\lambda+2)$), with Nariai $= (t-1)\times$clock. **(B) Relocation theorem.** The one missing object of the seam variational principle — the quantum clock — *cannot* live at the cosmological Nariai: the one-loop correction $\varepsilon = (c/24\pi)\Lambda/\bar{M}_{\mathrm{Pl}}^2 = 7.6 \times 10^{-122}$ (c=8, frozen ρ_Λ) is deficient by 121.5 orders, and the deficit *is* the Λ hierarchy $2\alpha^{-1}/\ln 10 = 119.0$. So the clock must live at the *seam* scale, where the only gapped generator is the established boundary transport — the bridge reaches its final form: *one* [C] identification, "the transfer operator is the seam's own Nariai clock". *(Since constructed, v124–v133: the resummed clock has the closed form* $\mathrm{rate}(n) = -p_2 \ln(1 - n/N_{\mathrm{fam}})$*, its weights are the Mehta–Seshadri parabolic weights of* $A_0^\star$*, and the rate is the entropy power law* $\Gamma \propto (S/S_{dS})^{p_2}$*; the residue of R1 is one finite* $\zeta(0)$ *budget.)* **(C) The complete gap list (machine-pinned).** Exactly *five* structural objects stand between the certified core and a strict TOE: the seam quantum clock [C], the holomorphy+c=8 theorem [C]/[O], the seam-determinant→EH step [O], the H2 dictionary [O]/[C], and the parabolic realisation of Q (which now also carries the sheet question) [C] — plus the two irreducibles (one scale v_{geo}, the primitive π) and the frozen data surfaces. Nothing is closed by this box; its claim is *completeness* — a sixth structural gap would fail the script.

Update (post v110–v122, [`verification/v123_inventory_update.py`*]):* the list keeps its cardinality but its *contents* sharpened. The old R4 ("H2 dictionary") is **discharged** — its values, addresses and margins are now exact theorems (v118–v122); what replaces it is the far smaller R4′, the *establishment of the selectors* ($n = (5, -9, 6)$ and the diamond determinants — five frozen integers). And R2 now carries **three loads**: the one index-4 boundary-net statement closes, when proven, the metric gate, the carrier choice *and* the QBL programme — one theorem, three doors. The flavor side has no open analysis class of its own anymore.

Second update (v141–v144): four of the five classes sharpened again, none removed. R5's *discrete* freedom is gone — the $\mathbb{Z}_3$ deck choice is *derived* from the integer model (geometric deck = sheet-twisted class, v141); R4′ shrinks from three pairings to *two* (the σ-pairing is integrality-forced mod 11, v142); R2's Q-system identification holds at the finite Lie level (graded Frobenius hull, v143) — the residue is *one* conformal-net statement; and R1's det-ratio cancellation is derived within the SdS family (e_2-rigidity, v144), with the finite-weight absorption obstruction stated sharply. R3 is untouched.

Third update (v145–v148): each class now sits at its floor. R4′: the pairing *values* are atom identities (new closure $|\mu_4| + g_{\mathrm{car}} = N_{\mathrm{fam}}^2$); residue = *one* lift map (v145). R5: the *full* Möbius D_4 of the seam curve matches the integer model parity by parity; residue = the module identification itself (v146). R1: the ring sum *is* the Born-squared Gaussian zero-mode integral and the quantum bend is det′-clean; residue = the measure identification + one reference normalisation (v147). R2: the odd glue sectors are *Ramond* modules (zero NS support), the glue choice is the R-projection ($248 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$) — the one net statement is scoped to the twisted-sector structure (v148). R3's mechanism now *exists* (gapped replica ⇒ EH form, v150), the Calderón transfer is *answered* (the DtN kernel is conically clean, BFK, v151), and its normalisation is the dimensionful anchor, not a separate gap (v152). And R4′ dissolves further: *both* dual normals are anchor data ($n = (p_2, p_0, e_2)(a)$ in the cusp frame, v149) — residue = one assignment.

Fourth update (v153–v154, external review formalised): the two surviving non-frontier items are now stated as theorems. v_{geo} is an *irreducible metrology primitive* by the **No-Unit Theorem** — a dimensionless compiler cannot select an absolute scale, and $U_{\mathrm{point}} \sim v_{\mathrm{geo}}$, $1/G \sim v_{\mathrm{geo}}^2$, $m/\mu = e^{3/4}$ are one unit in three mass-dimensions (v153). And G_{net} is the **Simple-Current Extension Theorem** $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$ (index 4, c=8, μ=1 ⇒ holomorphic ⇒ E_8), exact and machine-assembled; the only residue is the seam-side identification (v154) — which itself reduces to *one* shared physical premise: the boundary marginal of a Gaussian bulk is automatically quasi-free (compression of a contraction, v155), so G_{net}'s analytic core is the *same* "seam is a free RP field" premise that the area law (v59) and R3 (v150/v151) already use — one premise, not three. And the target net is *explicitly constructed and verified* (v156): the DtN/Calderón operator is the free chiral dispersion $\omega_k = |k|$, 16 Majoranas give c=8=5+3, the currents are 248=120+128, and the holomorphic character $E_4/\eta^8 = j^{1/3}$

(with 248 at level 1) makes $B \cong (E_8)_1$ a checked identity. And the free-bulk premise itself reframes (v157): freeness is the *rigid* holomorphic c=8 boundary fixed point — the DtN symbol is universally $|k|$ (Lee–Uhlmann), a holomorphic c=8 theory has no marginal $(1,1)$ deformation (so no room for an interaction), and the same $|\mathbb{Z}_2|$ that halves 8π in c_3 is the Ramond projection giving μ=1. So premise (A) shrinks from "prove Gaussianity" to "the gapped flow reaches the holomorphic fixed point"; freeness is then forced by rigidity, not postulated. An exact operator-dimension count (v158) settles the destination: the relevant window $(0,1)$ for bosonic deformations is empty, the dimension-1 currents are chiral, and the lowest interaction (a quartic) has dimension 2 (irrelevant) — the free chiral c=8 fixed point is isolated and stable. So (A) reduces to a basin-of-attraction statement with a proven stable target. *Fifth update (v160–v165): (A) ceases to be its own gate.* It is recast as a fixed-point theorem — quasi-free ⇒ all higher connected cumulants κ_{2n}=0 (v160); the infinite Schwinger cone is *eliminated*, because a quasi-free transport is the Bogoliubov second quantisation $\Gamma(t)$ and the cone gap equals the one-particle gap $(2/3)^6$ exactly (v161), a value forced by μ_4-equivariance and the unique residue $A_0^\star$ (v162); and the surviving 16-dimensional identification *factors* into the two already-open items — A_2 net existence (an assembly of established net constructions, [C]) and R_1 = `GATE.QGEO` (the seam collar boundary = the punctured sphere $\mathbb{P} \setminus \mu_4$) — so by the No-Unit Theorem the irreducibles stay exactly $\{\pi, v_{\text{geo}}\}$ and (A) carries *zero independent open gates* (v165); a unification, not an unconditional proof. **The honest one line:** TFPT is closed as a dimensionless boundary compiler; its only irreducibles are π and the one metrology unit v_{geo}; the one strict-TOE theorem still carrying weight is the seam-net identification; every "frontier" item is an instance of F_{transfer}.

Terminal inventory (machine-pinned, [`verification/v167_inventory_post_closure.py`]): with (A) discharged, the **(C)** count above collapses — the seam clock, the index-4 net and the Q-realisation all merge into the *one* Tier-1 hinge (seam = flavour = horizon = A_2 net assembly + `GATE.QGEO`), the EH step folds into v_{geo} and the ambient measure into G_{net}. The terminal residual is therefore *one* hinge + *one* transfer functor (F_{transfer}: Koide, η_B, axion, m_p/m_e) + the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem). A sixth independent structural class — or a new irreducible — would fail the script.

The carrier chapter: the interior is free, the structure is the certificate [E]/[C] ([`verification/v113_quasifree_kernel.py`])

The QBL round (v108–v113) adds the missing chapter of the origin story — *why this carrier*. The seam owns exactly *one* measuring device: a single scalar two-point kernel. Four theorems pin what such a device can do. *(i)* A scalar kernel exists *iff* it pairs the two sheets; there are exactly $2 = |\mathbb{Z}_2|$ such kernels — the glue ambiguity, resolved by the same sheet choice as everywhere ([`verification/v110_calderon_sheet.py`]). *(ii)* The certified channel *counts the code by itself*: one neutral kernel per code state, graded $(1,5,10)$ — the Pascal closure is two countings of one set, not a condition ([`verification/v112_selfcounting_channel.py`]). *(iii)* Pair transport is minimally complete: degree ≤ 1 generates nothing, degree 2 generates every code operation ([`verification/v111_quadratic_transport.py`]). *(iv)* The carrier net is 16 *free* Majorana fermions whose tower carrier $\to SO(16)_1 \to (E_8)_1$ never changes the field content — only the certificate grows (index $2{\times}2 = 4 = |\mu_4|$), and the central charge *is* the rank of the one kernel (5 carrier block, 8 seam hull). In the origin-story voice: **the interior of the world is free and structureless; everything we call structure lives in the certificate the seam issues** — E_8 is that certificate's unique checksum, the anchor $a = (1,1,2)$ its discrete genome, and the black-hole keystones (Nariai = anchor, one orientation, one clock) are the same bookkeeping read at the horizon. *Honest residue:* the premise "the seam is the free c=8 net" is the G_{net} gate itself — one theorem now closes both the metric story and the carrier choice — and that premise is itself no longer free-standing: it is a fixed-point theorem whose only residual factors into the already-open net-existence (A_2) and `GATE.QGEO`, with the irreducible core $\{\pi, v_{\text{geo}}\}$ a theorem (v160–v165). That residual is since sharpened to a single point: net existence and full-cone reflection positivity are discharged to [E] (v175), the realisation is assembled as one central theorem (v176) and split into the marks + kernel obligations whose finite cores are closed (v177/v178); the two then unify into one conformal-realisation premise (v179) which finally reduces — via uniformisation, Kerékjártó and the order-4 Möbius classification — to the milder, non-circular seam-*isometry* premise `QGEO.ISO.01` (the carrier clock is an order-4 orientation-preserving isometry of the RP seam metric, v180), and finally — by equivariant uniformisation — to the strictly weaker conformal-*symmetry* / deck premise `QGEO.SYM.01` (v181). Below that the residual is *definitional*: the carrier μ_4 clock *is* the conformal deck of the seam (the seam's conformal structure is the μ_4-deck structure of the carrier). It is since driven to its sharpest, *non-circular* form (v194–v198): the raw seam DtN is RP-canonical

(Osterwalder–Schrader on the quasi-free state), and the bedrock is cracked to the *state-invariance* $\omega \circ \rho = \omega$ — the DtN principal symbol commutes *exactly* with the clock, and Tomita–Takesaki gives $[\rho, \Lambda_\Sigma] = 0$ with no conformal-covariance assumption, removing the Bisognano–Wichmann circularity — the maximally-operational form of the same postulate. The bounded residual is then reduced once more (v201): writing the DtN as $\Lambda = |k| + M_f$ with M_f multiplication by the curvature $f(\theta)$, block-diagonality holds iff f is $\mathbb{Z}_4$-invariant, and a curvature sourced by the four μ_4 marks $f = \sum_j g(\theta - 2\pi j/4)$ *is* automatically $\mathbb{Z}_4$-invariant — so the marks being a μ_4 orbit (forced by v195) *force* the sub-principal symbol block-diagonal. The whole residual thus collapses to the *mark-locality* of the DtN (the seam flat away from the marks = the conformal-deck structure): a single definitional identification — not a finite computation, not reducible further without relabeling — left honestly **[O]**.

The deductive chain below the postulate is now machine-checked. The geometric normal form (Uniform: $z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, orbit$\to$ μ_4, multiplier order-4 $\Leftrightarrow$ $\zeta = \pm\, i$; `FORM.QGEO.02`) and the conditional implication *mark-local*$\Rightarrow$ $\omega \circ \rho = \omega$ (`FORM.QGEO.01`) are **Lean-formalised** (audit: pass, only the three standard kernel axioms; mirrors v177 and v201/v210). So everything *downstream* of the postulate is **[E]**; the postulate `QGEO.SYM.01` itself remains the one irreducible **[O]**.

Appendix: computational verification

Every **[E]** statement here is machine-checked in the TFPT suite and re-derived independently in Wolfram:

script	content
`v53_compiler_core.py`	$(5, 3)$ skeleton, Pythagorean Δ_Y split, anchor char-poly uniqueness
`v54_seam_horizon_keystones.py`	triply-forced 8; shared transport $\lambda_2{=}(2/3)^6$; de Sitter constants
`v55_coxeter_cycle.py`	computed E_8 Coxeter element (order 30), exponents = totatives, $S_{dS}\rho_\Lambda{=}32\pi^4$
`v56_unique_attractor.py`	gapped transport $\Rightarrow$ unique attractor, rate $(2/3)^6$; Coxeter planes; rank-1 reset

Run `python verification/run_all.py` (all pass) and, for the independent second path, `wolframscript -file verification/wolfram/tfpt_readouts.wl` (all pass). Figures are produced by `verification/make_figures.py`.